Richmond's Conclusions

Richmond et al. (2020) — The Human Standard

What the human review team did, what they produced, and what they concluded — stated plainly and grounded sentence-by-sentence in the published paper. This file is the reference that every system output is compared against. Nothing here is our interpretation unless marked *[our note]*.

Source: Richmond A, Cooper N, Gay S, Atiomo W, Patel R. *The student is key: A realist review of educational interventions to develop analytical and non-analytical clinical reasoning ability.* Medical Education. 2020;54:709–719. https://doi.org/10.1111/medu.14137 (open access, CC BY-NC-ND). Local copy: `data/paper-Richmond-original.pdf` (12 PDF pages = printed pp. 709–719 plus a copyright page). Page numbers below are printed page numbers. Figures 1–3 have no text layer in the PDF; they were transcribed from the rendered page images (PDF pages 3, 5 and 7).

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1. What the team set out to do (pp. 709–710)

Aim: *"to develop understanding about the way educational interventions develop effective analytical and non-analytical clinical reasoning ability, when they do, for whom and in what circumstances."* (Abstract, p. 709)

Research questions (p. 710): 1. *"What educational interventions are effective for developing analytical and non-analytical clinical reasoning ability amongst medical students?"* 2. *"When and why are they effective, for whom and in what circumstances?"*

Scope: undergraduate medical or health-care professions education (p. 711).

Background the questions rest on (p. 710): two complementary reasoning modes — **non-analytical** (fast, unconscious, triggered by familiarity, no effortful working-memory use) and **analytical** (effortful deliberation when a presentation is unfamiliar). Experts mostly reason non-analytically but switch to analytical reasoning when "things do not fit". Most existing interventions teach analytical processes and bias awareness; the evidence for them is "inconclusive"; interventions that build knowledge alongside analytical reasoning have shown benefit, but "not … for all learners, or across different reasoning tasks". Hence the realist question: why, for whom, in what circumstances.

2. The analytic framework they used (pp. 710, 712)

- A **realist review** builds a **programme theory** from **context–mechanism–outcome configurations (CMOCs)**. Context is *"separate to the intervention … but affect[s] how the intervention is received"*; it is not neutral.
- **Mechanisms are conceptualised as resources and responses**: *"interventions offer resources into a context, effecting a change in participants' responses, which in turn leads to various*

outcomes." (after Dalkin et al.). Responses (cognitive or emotional) are often invisible and *"can be theorised from existing theories of learning or inferred from data."*

- **Outcomes used**: *"diagnostic accuracy or effective learning about clinical reasoning as a process (eg, knowledge of case exemplars or illness script formation)"*, both short-term (diagnostic accuracy or error) and longer-term (development of process components over time).
- The review focused on **theory building rather than testing**, because interventions or outcomes of interest were often under-reported or too distant (p. 712).
- Standards: registered on PROSPERO (CRD42017072029); RAMESES publication standards for realist synthesis "were referred to throughout" (p. 710).

[our note] Our ontology mirrors this: Context, Intervention, Mechanism_Resource, Mechanism_Response, Outcome. Figures 2–3 label the response column "MECHANISM REACTION"; the text uses "Mresponse". Same concept.

3. How they did it — the human workflow, step by step

3.1 Initial programme theory (pp. 710–711)

1. The lead author (AR) ran background literature searches on clinical-reasoning interventions and dual-process theory.
2. All five team members (AR, NC, SG, WA, RP) were clinical teachers with expertise in clinical reasoning education. AR **built consensus** in the team about which interventions promote non-analytical versus analytical processes.
3. *"The outputs from the background literature search and consensus building led to the development of initial drafts of CMOCs. Following feedback from the research team, an initial programme theory (IPT) was constructed from these drafts."*
4. Two further theories were integrated into the IPT: **cognitive flexibility theory** (learning is context-dependent; transfer needs multiple scenarios) and **situativity theory** (learning is "situated in experience", with distinct contextual levels). *"These insights directed searching for contexts at the level of the student, the teacher and learning encounter or activity."*

3.2 Search (p. 712; Figure 1, p. 711)

- Databases: **MEDLINE (n = 3066), PsycINFO (n = 2779), ERIC (n = 1037), CINAHL (n = 745)**, searched in **May 2017**, from the **year 2000** onward (after *To Err Is Human*, 1999), using key themes developed from the IPT (Appendix S1).
- A **supplementary search** added terms that early searching showed relevant to expertise and non-analytical reasoning: *pattern recognition, deliberate practice, illness scripts, knowledge acquisition and recall.*
- "Knowledge" was defined multidimensionally (Table 1, transcribed in §4.2).

3.3 Screening and inclusion (p. 712; Figure 1)

- Duplicates removed: 530. **Titles and abstracts screened: 7097**; removed as irrelevant: 6958; added from the supplementary search: 10.

- **Full texts retrieved and reviewed for eligibility: 149.** Removed for one or more reasons (n = 124): unable to obtain full text 11; neurochemical/neurostructural aspects only 1; did not describe an *educational* intervention 59; did not specifically target dual-process theory 64; *assessment* of reasoning only 1; postgraduate only with no specific dual-process intervention 12; decision-support tools/checklists only 5; other 7; removed during data extraction as lacking methodological rigour 2; removed during data extraction as not contributing to the developing theory 2.
- Added from the reference lists of eligible full texts: 2. Added from a search on learning strategies to promote knowledge retention: 1.
- **Included in analysis: 28** (text: "25 articles met the inclusion criteria, and two more were added from reference list searching. One was added from a subsequent search …").
- Screening rules: the title/abstract screen "assessed for relevance … and studies [were] retrieved if they were deemed to **contribute to theory building**"; methodological rigour ("were the methods credible and trustworthy") was assessed at full text. An educational intervention was defined as *"a teaching process or method for developing knowledge and skills or delivering information to an individual or group."* Experimental studies that promoted a reasoning style by **giving instructions at the time of the test** were included as techniques that could be integrated into an intervention.

3.4 Extraction, analysis and synthesis (p. 712)

- *"The CMOCs were devised for all included full texts. Initial CMOC coding was undertaken by AR and all 28 articles were checked for consistency by another reviewer (RP, SG or NC)."*
- Data extraction forms were kept for all reviewed full texts (Appendix S1).
- *"Comparisons were made between studies and recurrent patterns of CMOCs were identified. Some studies particularly highlighted contexts whereas others shed more light on mechanisms. Studies identified earlier were re-analysed in light of theories arising from papers included later."*
- Tools: **NVivo 12** to store full texts and code contexts; **Microsoft Excel** to elaborate which contexts affected which mechanisms and outcomes.
- *"Key contexts and mechanisms for determining effectiveness were eventually produced as outputs from the synthesis through this iterative process."*
- Roles (Author contributions, p. 717): AR developed the protocol, performed the searches, initial extraction, synthesis, the visual representations and the first draft; RP, SG and NC reviewed a sample of full texts and gave feedback on the developing theory; AR, WA and RP conceived the study and the IPT.

[our note] The operations our agents must mimic are therefore: (1) seed a theory from background literature plus expert consensus; (2) theory-driven search with a supplementary term expansion; (3) two-stage screening by *contribution to theory* and then rigour; (4) lead-coder CMOC extraction with a second-reviewer consistency check on every paper; (5) cross-study comparison for recurrent patterns, re-analysing earlier papers under later theory; (6) synthesis into "key contexts and mechanisms for determining effectiveness".

4. What they produced — the outputs

4.1 Study characteristics (p. 712)

28 papers. Countries: UK 3, Canada 11, USA 3, Germany 2, Netherlands 3, other 6. Disciplines: medical education 23, veterinary education 1, psychology students 4. Total participants 1495. Outcomes were heterogeneous: diagnostic accuracy was the primary endpoint in 13 studies; 7 reported student satisfaction; 7 a change in knowledge; 5 described interventions theorised to promote illness-script and non-analytical reasoning development; 1 reported no outcomes but contributed to understanding mechanisms (Linn et al., 2012).

4.2 Table 1 — Definition of knowledge (p. 712)

TERM	DEFINITION
Knowledge	Encompasses factual, conceptual, procedural and metacognitive knowledge
Low knowledge	Generic low knowledge across all clinical domains
Low clinical domain-specific knowledge	Low knowledge within a clinical domain, low problem-specific knowledge or low knowledge specific to the clinical case
Inability to apply knowledge in a reasoning situation	Has sufficient generic or clinical domain-specific knowledge but lacks ability to apply this to a clinical reasoning case in a real or simulated clinical situation
High clinical domain-specific knowledge	Sufficiently high knowledge within a clinical domain or high problem- or clinical-case-specific knowledge
Different levels of knowledge	Differing generic levels of knowledge or problem-specific knowledge between students in the same group

4.3 The five student contexts (pp. 712–714) — the review's organising result

"The selected studies identified key contexts for clinical reasoning interventions at the individual level, with contexts at the teacher or the wider organisation level rarely discussed." Five key contexts: - **(a) students with 'low knowledge', low clinical domain-specific knowledge, or an inability to use knowledge in a reasoning situation** - **(b) students with high clinical domain-specific knowledge** - **(c) positive student coping strategies or appropriate level of self-confidence or self-efficacy** - **(d) negative student coping strategies or lacking self-confidence or self-efficacy** - **(e) students with different levels of knowledge within a group**

4.4 Figure 2 — "Individual student contexts important to the outcomes of teaching" (p. 713)

Transcribed row by row (Context → Mechanism resource → Mechanism reaction → Outcome):

CONTEXT	MECHANISM RESOURCE	MECHANISM REACTION	OUTCOME
Low knowledge / low clinical domain-specific knowledge / inability to use knowledge	Multiple relevant resources (see Figure 3)	Multiple relevant reactions (see Figure 3)	Positive and negative outcomes depending on mechanisms

CONTEXT	MECHANISM RESOURCE	MECHANISM REACTION	OUTCOME
High clinical domain-specific knowledge student	Instructing the use of analytical reasoning alone, especially with low-difficulty cases; teaching strategies that promote 'overthinking'	Able to trust 'sense of familiarity' and maintain diagnostic accuracy	No improvement in diagnostic accuracy. Limited or no increase in learning gain or outcomes
High clinical domain-specific knowledge student	Reasoning processes and outputs identified and discussed with facilitator	New understanding about the process of clinical reasoning	Positive impact on learning outcomes. Consolidates understanding about the process of clinical reasoning
Positive coping / appropriate self-confidence or self-efficacy	Simulated environments that promote authentic real-life situations, or teaching that enables making mistakes, or teaching in the real world	New understanding, learners 'feel safe' to make mistakes	Positive impact on learning outcomes and building or refinement of 'illness scripts'
Negative coping / lacking self-confidence or self-efficacy	*(the same resource as the row above)*	Fear, stress or anxiety	Negative impact on learning outcomes and building or refinement of 'illness scripts'
Different levels of knowledge within a group	Teaching approach designed around effective learning strategies to increase long-term knowledge retention; comprehensive feedback received in a timely manner following a reasoning task	Continued development and understanding about the process of clinical reasoning	Positive impact on learning outcomes, development of non-analytical knowledge, and building or refinement of 'illness scripts'
Different levels of knowledge within a group	Insufficient or incomplete feedback (including incorrect or erroneous in nature)	Confusion	Negative impact on learning outcomes

4.5 The CMOC statements, verbatim, with the studies behind each (pp. 714–716)

Context 1 — low knowledge, low clinical domain-specific knowledge, or inability to use knowledge (22 studies)

> *When students have low knowledge, low domain-specific knowledge or an inability to apply their knowledge in a reasoning situation (C), there are multiple ways (Mresource) in which educational interventions may develop their diagnostic accuracy or reasoning ability (O). The context of low knowledge (C) combines with different resources to produce varying emotional and cognitive responses in the students (Mresponse), which promote either positive or negative educational outcomes (O).*

Explanation given: this context "exerted most influence on the interventions"; the many pathways are in Figure 3 (§4.6). The text's own example: students with insufficient knowledge who **passively observe an expert without an explanation** "may experience panic or resentment at not immediately knowing how the expert has made a decision"; students who **receive a full and explicit explanation of an expert's reasoning** "tend to have positive learning experiences". Many undergraduates are novices in most clinical situations; a student can have "adequate knowledge" for their stage yet "low knowledge" relative to the challenge, or be perceived as low-knowledge when unable to apply it. Studies (Richmond refs 9, 26–31, 35–49): Chamberland 2013; Kulatunga-Moruzi 2001; Ark 2006; Eva 2007; Ark 2007; Myung 2013; Evered 2013; Linn 2012; Bender &

Danielson 2011; Graafland 2014; Nendaz 2011; Borleffs 2003; Klamen & Williams 2011; Jacobson 2010; Mamede 2012; Blissett 2012; Chamberland 2011; Coderre 2010; Pinnock 2012; Adams 2011; Chamberland 2015 (Med Educ); Chamberland 2015 (Adv Health Sci Educ).

Context 2 — high clinical domain-specific knowledge (7 studies)

> *When an expert's reasoning processes or thoughts are explicitly revealed and discussed (Mresource) with students with sufficient domain-specific knowledge (C), this promotes understanding (Mresponse) leading to insight (Mresponse) into the reasoning process when diagnosing and managing patients (O) and a positive learning experience (O).*
> *When students with high clinical domain-specific knowledge (C) are instructed to use analytical reasoning alone or this is promoted by ensuring they think through all aspects of the case (Mresource), they may feel frustrated (Mresponse) as they can rely on non-analytical reasoning (Mresponse) and still attain high diagnostic accuracy (O).*

Explanation: sufficient knowledge and experience enable relatively accurate non-analytical reasoning; such students "may benefit more from discussions with teachers and experts explaining their reasoning", and "derive little benefit from being stretched for explanations, or when directive teaching approaches, commonly reserved for 'low knowledge' students, are used on them." Studies (refs 8, 9, 30, 38, 44, 50, 51): Sherbino 2014; Chamberland 2013; Myung 2013; Nendaz 2011; Chamberland 2011; McGregor 2012; Bloice 2014 (plus the expertise-development literature).

Context 3 — positive coping strategies or appropriate self-confidence/self-efficacy (5 studies)

> *When students with the ability to cope with the challenge of performing clinical reasoning (C) or students with an appropriate level of self-confidence or self-efficacy calibrated to previous clinical reasoning performance (C) are exposed to teaching resources that allow them to make mistakes (Mresource) or those based in real-life scenarios (Mresource), including simulation and simulated patients, they feel grateful for the learning experience (Mresponse). The experience enables them to build understanding (Mresponse), which has a positive impact on learning (O) and this is important to developing more complete illness scripts and performing more accurate non-analytical reasoning (O).*
> *When students with positive coping strategies (C) are exposed to real cases (Mresource) they feel pressure that their decision making could have a real impact (Mresponse), which results in a positive learning experience (O).*

Explanation: stress in simulated or real environments can be intrinsic or extrinsic; emotions perceived as necessary for the task may enhance knowledge acquisition, those perceived as peripheral impair performance; whether triggers are seen as threats or opportunities depends on coping strategies; "experience, practice and making mistakes are vital". Studies (refs 36, 37, 46, 50, 51): Bender & Danielson 2011; Graafland 2014; Pinnock 2012; McGregor 2012; Bloice 2014.

Context 4 — negative coping strategies or lacking self-confidence/self-efficacy (4 studies)

> *Students with poor coping strategies or low self-confidence or self-efficacy beliefs calibrated to previous clinical reasoning performance (C), exposed to simulated or real patient encounters (Mresources), may experience fear (Mresponse), stress (Mresponse) or pressure to perform (Mresponse). As a consequence, cognitive load is increased (Mresponse) resulting in poor illness script development, faulty future non-analytical reasoning and negative learning outcomes (O).*

Explanation: negative outcomes result from inability to cope with the stress of simulated or real environments and with making mistakes; simulation lets students make mistakes without fear of punishment, yet "not all students will make learning gains in these situations." Draws on cognitive load theory and human stress responses. Studies (refs 36, 46, 50, 51): Bender & Danielson 2011; Pinnock 2012; McGregor 2012; Bloice 2014.

Context 5 — different levels of knowledge within a group (9 studies)

> *When teaching students with different levels of pre-existing knowledge in a group (C), using strategies that promote knowledge retention (Mresource) will build upon what they already know (Mresponse) and develop understanding (Mresponse) regardless of their pre-existing knowledge level, which leads to increased learning (O) and further engagement (O). Providing accurate and timely feedback (Mresource) is an important component of developing reasoning skills in students with all levels of knowledge (C), to develop understanding (Mresponse) of their successes and failures and generate plans for improvement. This is more likely to promote development of complete illness scripts (Mresponse) and successful non-analytical reasoning in the future (O). When feedback is absent, incomplete or contains errors (Mresource) this can lead to confusion (Mresponse) and have a negative impact on learning (O).*

Explanation: the challenge is teaching students with varying pre-existing knowledge; the most effective approach is timely, accurate feedback plus strategies for long-term retention; one effective strategy was **test-enhanced learning**; absent or erroneous feedback "led to confusion, reduced understanding and impaired clinical reasoning development." Studies (refs 29, 40, 43, 46, 47, 51, 56–58): Ark 2007; Klamen & Williams 2011; Blissett 2012; Pinnock 2012; Adams 2011; Bloice 2014; Kopp 2008; Boshuizen 2012; Raupach 2016.

4.6 Figure 3 — CMOCs for the low-knowledge context (p. 715)

Context for every row: *students with 'low knowledge', low clinical domain-specific knowledge, or an inability to use knowledge in a reasoning situation.*

Facilitators → outcome: increase in learning gain or outcomes, or increase in diagnostic accuracy | Mechanism resource | Mechanism reaction | |---|---| | Listen to near-peer 'think aloud' their reasoning with the use of prompts and examples | Assume similar level prior knowledge resulting in sense of 'feeling at ease' | | Instructions to use both 'non-analytical' or *pattern recognition* and analytical or step-wise approach to reasoning | Empowered to trust in their 'sense of familiarity' and developing ability | | Accurate feedback in a timely manner **and** explicit and clear explanation of expert's reasoning | Provides a sense of clarity and helps develop understanding. Affirmation the individual possesses sufficient knowledge for reasoning | | Promotion of analytical

or step-wise approach as a reasoning scaffold | Relieves the tension of not immediately knowing the answer. The learner feels comfortable, supported and at ease |

Barriers → outcome: decrease in learning gain or outcomes, or decrease in diagnostic accuracy | Mechanism resource | Mechanism reaction | |---|---| | Inducing or imposing time constraint to deliberately force non-analytical reasoning | Spontaneous outputs from non-analytical reasoning (including *guessing*) with resultant frustration and potential distress | | Passive observation of experts without receiving explanation about their reasoning processes | Resentment or panic at not recalling or 'knowing' immediately | | Listening to experts explain their reasoning, which may skip steps or use pattern recognition | Find it difficult to understand the non-analytical thought processes, discordance between illness scripts | | Increased case difficulty or case with significant reasoning challenge | Frustration | | Listening to peer reasoning as a passive recipient when that self-explanation includes mistakes | Confusion |

4.7 What the Discussion adds (pp. 716–717)

- Interventions *"do not work independently, but work inter-dependently with the individual, their pre-existing clinical domain-specific knowledge and their ability to cope"*; this is the **first study to report that the same educational intervention may 'work' across many contexts, but the effect is not the same across them all, especially amongst students with low knowledge states**; a one-size-fits-all approach is less effective; negative outcomes occur when students are given tasks beyond their competence "even with a teacher present".
- **Matthew effect**: better pre-existing knowledge correlates with higher achievement, so identifying a student's knowledge level for a task or domain is necessary to predict which interventions will be effective.
- **Expertise reversal effect**: more experienced learners may not benefit, or may reverse gains, from techniques designed for novices.
- Time spent on a course is a poor marker of competence; improving knowledge structures and long-term retention is essential for both confidence and competence; low-knowledge students "need support the most".
- Recommended learning strategies: retrieval practice, interleaving and spaced practice ("desirable difficulties"), ideally before or alongside simulated or real experiences.
- Exposure to real patients (in practice or simulation) is necessary for "safe mistakes" and for developing coping strategies; self-management of the emotions evoked by reasoning matters.
- Interventions have favoured analytical processing; students with sufficient knowledge should be **encouraged toward non-analytical reasoning**, switching back to analytical reasoning under complexity or uncertainty. Educators should identify baseline knowledge, monitor development over time, and feed back which approaches need improvement.
- Strengths and limitations: the first realist review of why these interventions work, for whom and in which contexts; iterative triangulation of insights; some outcomes are not explicitly reported by primary authors; better reporting standards would improve future syntheses.

4.8 Conclusions (p. 717, verbatim)

> *Educational interventions for developing analytical and non-analytical reasoning amongst undergraduate students in medical or health care professions education predominantly work by increase knowledge acquisition, mobilisation and recall alongside encouraging practice in real or simulated environments. Students with low knowledge states affect more whether educational interventions are successful or not, therefore identifying what individuals know about a particular clinical problem or across a knowledge domain is important. Students with developing expertise should be encouraged to use non-analytical reasoning just like experts do, switching back to analytical reasoning when faced with complexity or uncertainty and needing to explain their thinking.*

Headline (title and abstract): **the student is key** — the key student-level factors are self-confidence, self-efficacy and pre-existing knowledge; these contexts influence interventions, "impacting both positively and negatively on educational outcomes."

5. Richmond's observable outputs, and what of ours is compared with each

#	RICHMOND OUTPUT (WHERE)	OUR CORRESPONDING OUTPUT	COMPARISON
1	Included set of 28 studies (Fig. 1; Appendix S2)	Screening decisions (`screening_decisions`)	inclusion agreement: sensitivity on the 28; specificity on a distractor pool
2	Five student contexts (p. 714; Fig. 2)	Context entities and the context sections of our programme theory	are the same five contexts present and populated?
3	Per-context CMOC statements (§3.2.1–3.2.5) and Figure 2/3 rows	CMOCs (`cmocs` , `entity_instances` , `typed_relations`) normalised to canonical concepts	concept recovery (E-codes), relation recovery (R-codes), configuration recovery per context
4	Per-context study membership (reference numbers per context, §4.5)	the studies whose CMOCs carry each Context entity	study-level agreement per context
5	Key mechanisms "for determining effectiveness" (pp. 712, 716)	Leiden big concepts and demi-regularities	community ↔ mechanism alignment
6	Programme theory as five context-specific chains, plus the Discussion claims (Matthew effect, expertise reversal, one-size-fits-all fails)	Programme theory (`programme_theories` , latest version)	structural/conceptual correspondence per chain; presence of the three headline claims
7	Verbatim-grounded claims (data extraction forms, Appendix S1)	every extracted element carries a verbatim quote resolved to a character span	citation faithfulness

The machine-readable form of §4.3–4.6 is `gold/richmond_gold.json` (E-codes, R-codes, PTS chains, context membership). The full verification design is in `docs/OUTPUT_AND_VERIFICATION.md` .

6. Corrections to earlier project documents (recorded for transparency)

- The previous version of this file stated that Figure 2 was the *initial* programme theory and Figure 3 the *refined* programme theory. That was wrong: Figure 2 shows the five student contexts with their mechanisms and outcomes; Figure 3 details the configurations for the low-knowledge context only (captions on pp. 713 and 715). The old file is kept as `gold/RICHMOND_CONCLUSIONS.v1-2026-07-15.md`.
- The v1 gold programme-theory chains PTS2 and PTS5 listed interventions that Richmond assigns to other contexts (PTS2 used E10/E11, but §3.2.2 names E07 and E08; PTS5 used E07/E17, but §3.2.5 names E12, E13 and E14). They are corrected in `gold/richmond_gold.json` v2.0; the v1 file is kept as `gold/richmond_gold_v1.json` so earlier scores remain reproducible.
- Nine Figure-3 resources and reactions had no E-code in v1; they are added in v2.0 as a clearly flagged supplement (E48–E59, R41–R61) so coverage can be reported against both the original 47 concepts and the complete figure.